Participatory /Speculative **Design**

UXpress Yourself
29 October 2026

Agenda

10m	Introductions
20m	*Tour de table* activity
30m	Intro to Participatory and Speculative Design
15m	Break
60m	Hands-on experience
45m	Role-Playing and Discussion



Discordo

Neutro

Concordo

Torrada e galão é um melhor pequeno-almoço do que tosta de abacate com ovo escalfado

Personas são uma perda de tempo

O livro é sempre melhor que o filme

A análise de entrevistas feita por IA é hoje tão boa como a de um investigador experiente

Não é sempre necessário seguir todos os passos nas instruções de montagem de um móvel do IKEA

Há ocasiões em que é OK usar Comic Sans para UI

Mensagens em audio não devem ultrapassar 30 segundos

Em 3 anos, a maioria dos interfaces vão ser gerados por IA, em tempo real para os utilizadores

Ouvir *audiobooks* conta como ler

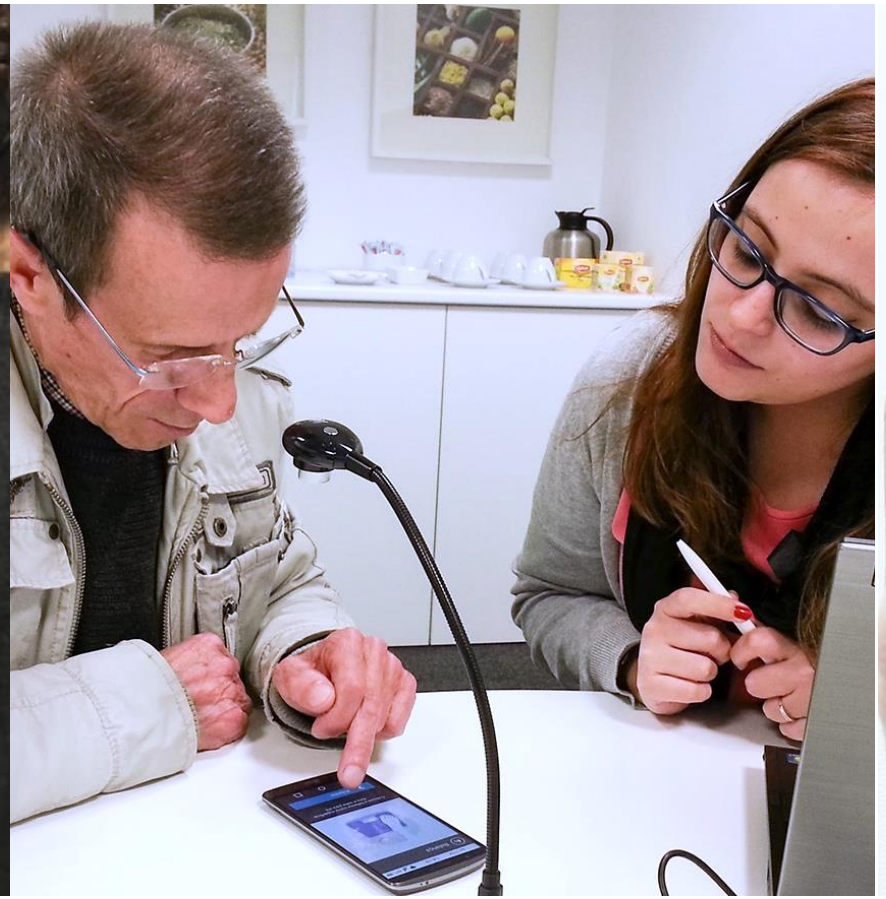
Todo o UI design é manipulação



Human-Centred Design



DESIGN RESEARCH



HUMAN-COMPUTER INTERACTION



HEALTH TECHNOLOGY ASSESSMENT AND IMPLEMENTATION


Participatory Design

1984 Designing *for* Consumers

2014 Designing *with* Users

2044 Design *by* People

Sanders, L., and Stappers, P. J. (2014). *From designing to co-designing to collective dreaming: three slices in time.* Interactions, 21 (6): 24–33

Participatory design is an approach where the people who are affected by a design are actively involved in the design process.

Schuler, D., & Namioka, A. (Eds.). (1993). *Participatory Design: Principles and Practices.* Lawrence Erlbaum Associates.



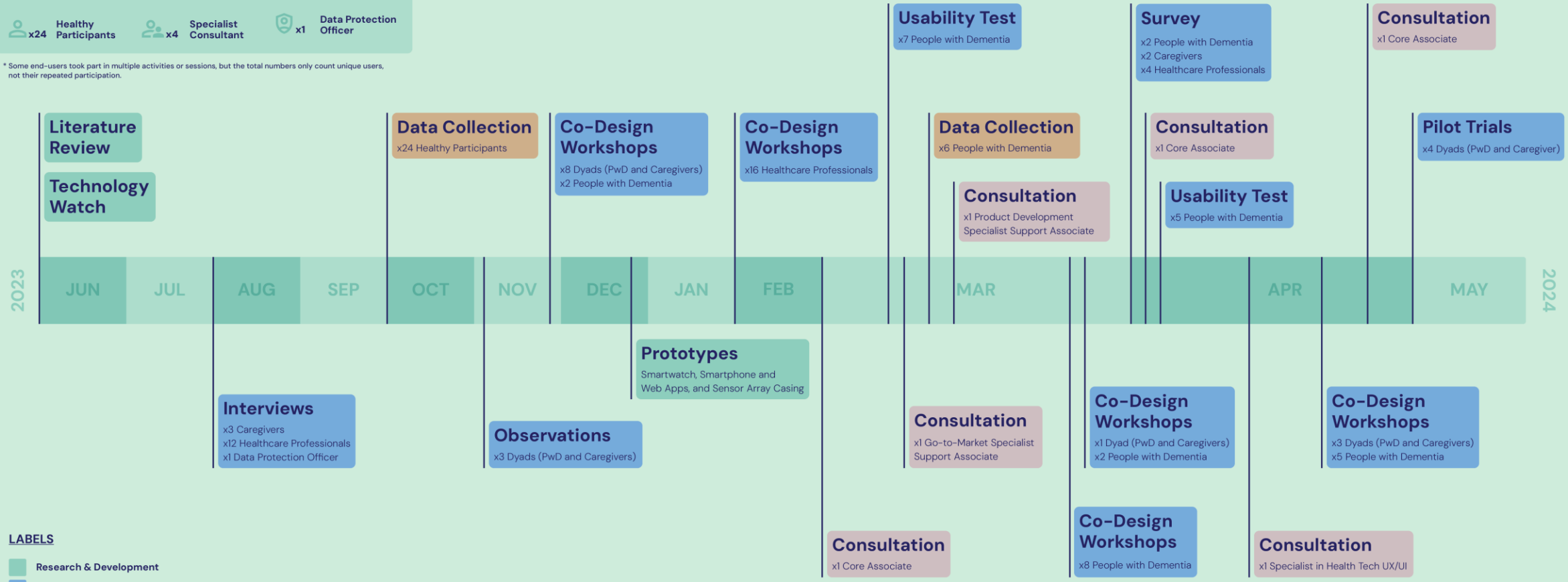

103
102

THE WORK

PARTICIPANTS *N*=121*

- x46 People with Dementia
- x18 Informal Caregivers
- x28 Healthcare Professionals
- x24 Healthy Participants
- x4 Specialist Consultant
- x1 Data Protection Officer

* Some end-users took part in multiple activities or sessions, but the total numbers only count unique users, not their repeated participation.



LABELS

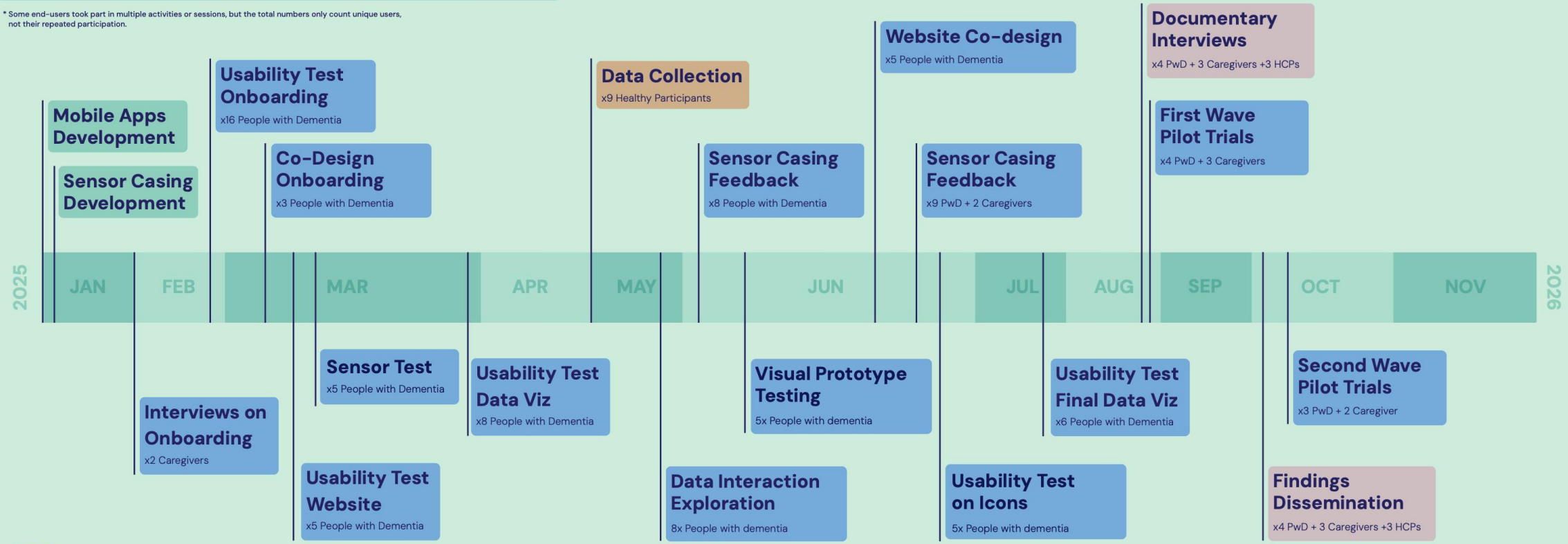
- Research & Development
- User Research, Participatory Design and User Testing
- Collect data to train ML models and algortithms
- Support from LPoD

THE WORK

PARTICIPANTS *N=64**

- x35 People with Dementia
- x10 Informal Caregivers
- x10 Healthcare Professionals
- x9 Healthy Participants

* Some end-users took part in multiple activities or sessions, but the total numbers only count unique users, not their repeated participation.



LABELS

- Research & Development
- User Research, Participatory Design and User Testing
- Collect data to train ML models and algortithms
- Dissemination

Specul at i ve Desi gn



“Priscila, 37 kilotons Nevada, 1957”
Huggable Atomic Mushroom
Dunne and Raby, with Anastassiades, 2004



"an activity where conjecture is as good as knowledge, where futuristic and alternative scenarios convey ideas, and where the goal is to emphasize implications of "mindless" decisions for mankind."

"Speculative Everything: Design, Fiction and Social Dreaming" Anthony Dunne and Fiona Raby



A/B Manifesto

Design Afirmativo e Design Crítico

A	B
Affirmative	Critical
Problem solving	Problem finding
Provides answers	Asks questions
Design for production	Design for debate
Design as solution	Design as medium
In the service of industry	In the service of society
Fictional functions	Functional fictions
For how the world is	For how the world could be
Change the world to suit us	Change us to suit the world
Science fiction	Social fiction
Futures	Parallel worlds

A	B
The “real” real	The “unreal” real
Narratives of production	Narratives of consumption
Applications	Implications
Fun	Humor
Innovation	Provocation
Concept design	Conceptual design
Consumer	Citizen
Makes us buy	Makes us think
Ergonomics	Rhetoric
User-friendliness	Ethics
Process	Authorship



Auger, J., & Loizeau, J. (2001). Audio Tooth Implant



2R THE SUN, Saturday, June 29, 2002 2E

RADIO CHEW

Listen to shows and phone via your tooth

Inventor . . . clever James

By PAUL CROSBIE

RADIO fans could soon be able to pick up programmes and take mobile phone calls via a gadget in their TOOTH.

The electronic device, including a receiver and a tiny vibrator, would be implanted in routine dental surgery.

It would pick up digital radio signals, then convert them to audio.

The sound would be transferred from the tooth to the inner ear by making the jaw bone vibrate.

Helped

It could allow the listener to receive information anywhere and at anytime.

So off-duty City traders could be kept up to date

remote. It is believed to be the world's first in-body consumer product.

The device was invented by James Auger, from Derby – helped by his pal Jimmy Loizeau, 23, from Rhyl, North Wales.

Royal College of Art graduate James, 23, said: "It's not science fiction, it's the stuff of science fact.

"It's fairly simple technol-

because we did not want to thrust this too far into the future to enable people to understand and believe it."

James and Jimmy demonstrated how the device would work at London's Science Museum yesterday.

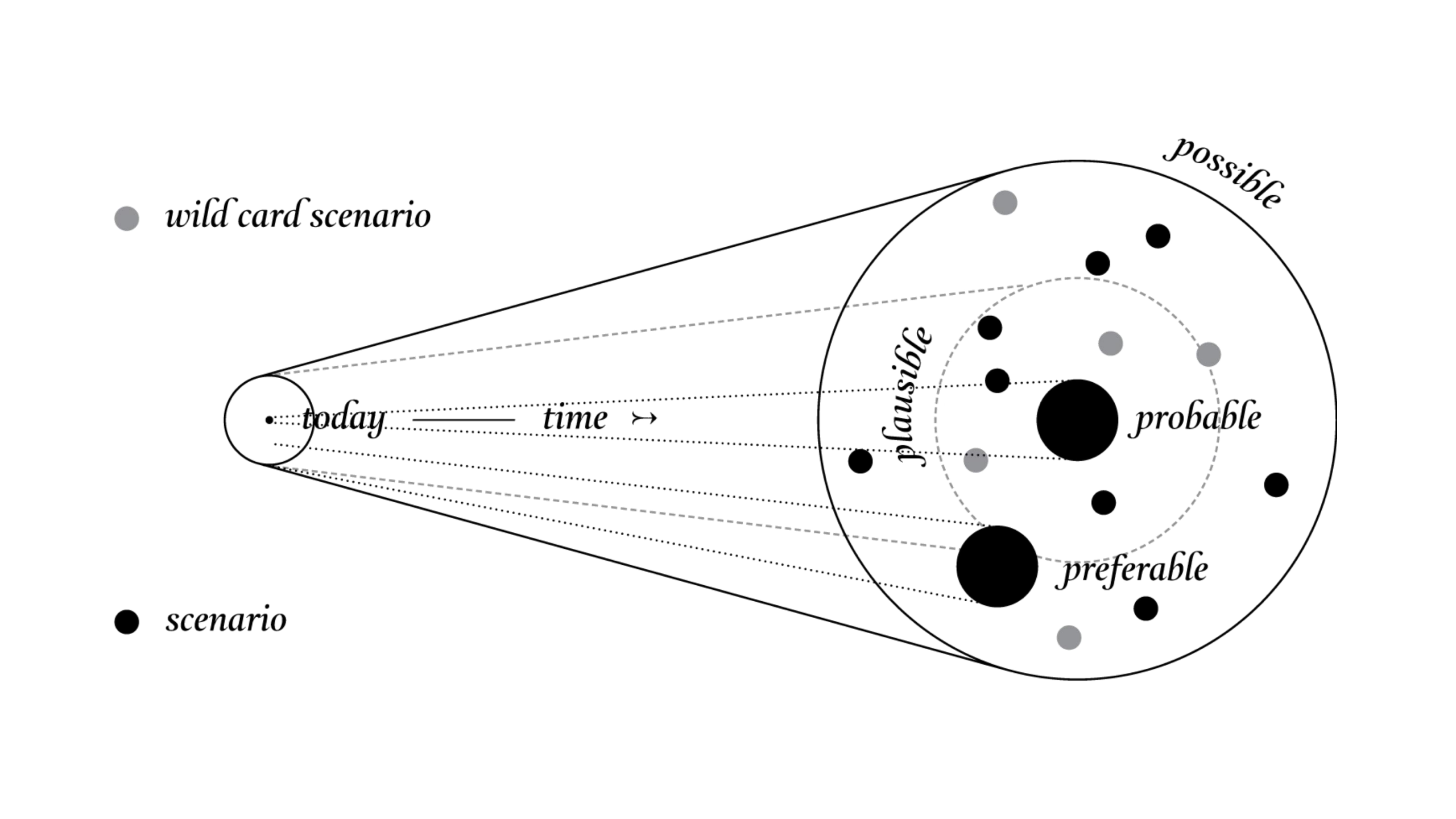
Museum visitors can also test the technology by using an adapted loudspeaker and a cocktail stirring stick.

By placing one end of the





Ai Hasegawa,
I Wanna Deliver a Shark, 2012


wild card scenario
scenario
today
time
possible
plausible
probable
preferable





Project Anathema

OPO Citizen's Office

Visual Health Assessment Scanner

You will undergo a Visual Health Assessment Scan. This examination is a diagnostic method that captures detailed images of your eye health without the use of radiation or invasive procedures and is designed to assess and monitor your visual condition effectively.

By signing and submitting this form, you agree to the terms and conditions outlined below, including data processing and usage policies.

PATIENT IDENTIFICATION

Full Name: _____
Adress: _____
Date of Birth (DD/MM/YYYY): __/__/__ Sex: ( ) Mal ( ) Fem ( ) Other _____
Contact Number: _____ Age: _____ Weight: _____
Email Address: _____ Height: _____

INTRODUCTION TO THE PROCEDURE

How is it performed?

You will undergo a biometric full-body scan using our advanced diagnostic imaging system. This scanner analyses detailed biometric data, including vital signs, metabolic rates, vision metrics, and other critical health indicators.

The scan is entirely non-invasive, relying on a combination of bio-optic sensors, advanced imaging techniques, and real-time data processing. During the procedure, you will lie comfortably on a motorized table that moves through the scanning device. The system is designed to ensure optimal comfort, with open ends, adjustable lighting, and constant airflow to prevent discomfort.

Will I receive any contrast or medication?

Depending on the examination requirements, a small amount of contrast agent may be administered to enhance the imaging of specific tissues or organ systems. This agent is designed to improve the detection and analysis of abnormalities.

Patients with known allergies to contrast substances, or specific medical conditions, such as kidney impairments, must inform the healthcare professional before the procedure. If contrast is required, it will be injected intravenously under close monitoring. Reactions to the contrast are extremely rare, but nausea, headaches, or mild site irritation may occur in isolated cases.

After the exam:
The patient may resume their normal routine without the need for rest or additional care, except in cases where anaesthesia was used or as directed by the physician.

PREPARATION

Why might there be delays in your exam schedule?
While we strive to adhere to scheduled times, unforeseen circumstances may arise, such as equipment malfunctions, system issues, power outages, medical emergencies, or the need to assist patients with higher dependency levels. If delays occur, we will take measures to ensure you are attended to with the utmost quality and care.

To ensure accurate results and your safety you may be asked to:

-Change into a provided gown to avoid interference from metallic objects or fabrics.

-All personal items, such as jewellery, watches, credit cards and electronic devices are to be stored securely before entering the scanner room, as the device uses advanced electromagnetic fields.

-Answer all pre-scan questions honestly, particularly regarding implants, pacemakers, or other medical devices that may interact with the scanner's technology.

During the scan, you must remain still to ensure high-quality results. The scanner may emit soft sounds, and you will be in constant communication with the operator through an intercom system.

Possible side effects and complications:
In rare cases, adverse reactions to the contrast agent may occur, such as discomfort, nausea, headache, hives, or, in extremely rare cases, severe allergic reactions. Our medical team is trained to respond promptly to such situations. More severe complications, such as joint infection, are extremely rare but should be reported to your doctor if symptoms like persistent pain, heat, or redness at the exam site occur.

Data collection and reporting:
During the exam, relevant health data will be collected and analysed to generate a detailed report, which will be sent to the physician responsible for your care.

Additionally, to support advancements in diagnostic and treatment technologies, anonymized data may be included in statistical databases and utilized in the development of machine learning algorithms. These technologies enhance diagnostic accuracy, optimize clinical processes, and contribute to medical innovation. Should you prefer not to participate in these initiatives, please inform our team prior to the exam.



Project Signo

EndoCards

A reflective card deck for expressing, understanding and honoring the experiences of people living with endometriosis



UTERUS CLOWN

APPEARANCE

It has tubes and ovaries.
It screams.

CHARACTERISTICS

Always interfering into life.
It is tricky.

CAPACITIES

Can be mild.

LIMITATIONS

Can be scary and bad.

It is similar to a joker.
It causes depression.
It screams and it is unpredictable.
From one moment to another it goes from a good clown to a bad, evil one.
It is the main character of the story, but should only be a supporting character.
It depends on other supporting characters to exist.



NO NAME

APPEARANCE

Sea mine and kind of a force of nature but living.
"Monster".

CHARACTERISTICS

Creeping in the shadows.
Ticking time bomb.
Spikes and tendrils that can reach far.
Takes over.
Can't be found.
Destruction.

CAPACITIES

Reach far.
Nothing can fight against it.
Consumes everything.

LIMITATIONS

None.

There's a notion of a feeling, a light feeling that there is someone in your body. Nobody else can find it. Suddenly, you feel an explosion, the impact, the burning of the insides. The pain reaches almost every part of your body at that time. You can feel the spikes trying to rip through your stomach, you can feel the mine's tendrils reach to your legs, your chest, your head. it consumes you and that is the moment you know, you will never have full control of your body, ever again. It got you, you are never free again, and there is nothing that can fight this force of nature.

Vaz-Pires, C. (2025). *Manifesting the Lived Experience of Endometriosis*


Project Fictions

Referências

Bibliografia e modelos conceptuais

- Sanders, E. B.-N., & Stappers, P. J. (2014). From designing to co-designing to collective dreaming: Three slices in time. Interactions, 21(6).
- Schuler, D., & Namioka, A. (Eds.). (1993). Participatory Design: Principles and Practices. Lawrence Erlbaum Associates.
- Dunne, A., & Raby, F. (2013). Speculative Everything: Design, Fiction, and Social Dreaming. MIT Press.
- Dunne, A., & Raby, F. (2009). A/B Manifesto.
- Voros, J. (2003). A generic foresight process framework. Foresight, 5(3), 10–21.

Projetos e estudos de caso

- Dunne, A., Raby, F., & Anastassiades, M. (2004). Priscila Huggable Atomic Mushroom. Designs for Fragile Personalities in Anxious Times.
- Auger, J., & Loizeau, J. (2001). Audio Tooth Implant.
- Hasegawa, A. (2012). I Wanna Deliver a Shark...

Projetos Fraunhofer AICOS

- AUTONOMOUS: An intelligent UbiquiTOus solutioN fOr MultimOdal daily living procedUreS tracking in people with dementia. [link]
- ANATHEMA: Technology for ageless sexual health. [link]
- Signo: Value systems in digital health technologies. [link]
- Vaz-Pires, C. (2025). *Manifesting the Lived Experience of Endometriosis* [MSc Erasmus Mundus Joint Master in Service Design Strategies and Innovations]. [link]
- FICTIONS: Co-creation of future visions for digital technology supporting manufacturing workers. [link]

Thanks!

ricardo.melo@fraunhofer.pt